

Mohamed Juma

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PROFESSIONAL SUMMARY

Full Stack Developer and Microbiology graduate building at the intersection of software, education technology, and cybersecurity. Currently developing EduTrak, a multi-tenant school management system for Kenya's CBC curriculum, and BioTrak, a production DNA sequence analysis platform integrating bioinformatics algorithms with modern web technologies. Seeking internship or junior developer roles in EdTech, cybersecurity-adjacent engineering, or developer tooling.

EDUCATION

Karatina University, Karatina, Kenya

Bachelor of Science in Microbiology

Graduated 2024

TECHNICAL SKILLS

Languages & Tools: JavaScript (ES6+), Python, Bash, SQL, Linux CLI, Git/GitHub

Frameworks & Libraries: React, Node.js, Express.js, FastAPI, Prisma ORM, ReportLab, Recharts

Databases: PostgreSQL, MongoDB

APIs & Auth: REST APIs, JWT, OAuth 2.0, Anthropic Claude API, NCBI E-utilities

Cybersecurity: Kali Linux, Nmap, Gobuster, SQLMap, HackTheBox, Proxychains, x64 Assembly

DevOps & Deployment: Docker, Docker Compose, Celery, Redis, Vercel, Render, Neon (PostgreSQL)

PROJECTS

BioTrak (*Production — Live*)

Stack: React, FastAPI, Python, PostgreSQL, Redis, Celery, ReportLab, Docker

Production DNA sequence analysis platform offering 10 scientific analysis modules in a single browser-based interface. Implements custom pure-Python algorithms for base composition, ORF detection across all six reading frames, nearest-neighbour primer design (SantaLucia, 1998), Smith-Waterman local alignment, restriction enzyme mapping (40 enzymes), GC skew analysis, RSCU codon usage scoring, and protein physicochemical property calculation. Features async NCBI BLAST integration via Celery workers, AI-powered sequence interpretation via the Anthropic Claude API, JWT-based authentication with role-based access control, and a ReportLab PDF report generator. Deployed on Render and Vercel using Neon PostgreSQL and Upstash Redis on free-tier infrastructure.

GitHub: github.com/modd3/BioTrak

EduTrak (In Development)

Stack: React, Express.js, PostgreSQL, Prisma ORM, JWT

Next-generation multi-tenant school management system supporting Kenya's CBC and 8-4-4 curricula. Multi-tenant architecture allows independent operation per institution, covering student records, fee tracking, grading, and report generation within a single platform.

GitHub: github.com/modd3/edutrak

School Management System (Live in Production)

Stack: MongoDB, Express.js, React, Node.js

Deployed school management platform handling student records, grading, and automated report generation. Reduced administrative workload by an estimated 70% for end users.

GitHub: github.com/modd3/school-management · Live: school-management-o50k.onrender.com

AI Prompt Builder (In Development)

Stack: MERN Stack, AI APIs, OAuth, REST API

Social platform for AI practitioners to create, test, version, and share prompts collaboratively. Supports community discovery of high-performing prompt patterns and iterative prompt development workflows.

GitHub: github.com/modd3/ai-prompt-builder

Proxy-Xtractor (Stable)

Stack: Bash, Python, Web Scraping, Kali Linux

Automated tool for penetration testers that scrapes and validates free SOCKS5 proxies then auto-configures proxychains4 for Kali Linux security workflows.

GitHub: github.com/modd3/proxy-xtractor

IP2Binary (Stable)

Stack: Bash, Python, Networking

CLI utility converting IPv4 addresses to binary and back for subnetting practice and network fundamentals study.

GitHub: github.com/modd3/ip2binary

CERTIFICATIONS

- HackTheBox — Active Member (2024–Present)
- freeCodeCamp — Responsive Web Design (2023)
- freeCodeCamp — JavaScript Algorithms & Data Structures (2023)

TARGET ROLES

Internship or Junior Full Stack Developer positions in EdTech, cybersecurity-adjacent engineering, or developer tooling. Available for remote work and open to relocation.